

Base architecture for the worker pipeline.

The pipeline should contain an always-on component that listens to requests, and a dynamic, scalable component that is triggered by the first one. This is required to satisfy both FR-4 and FR-11.

The always-on component consists of an endpoint that accepts the name of a state, and allocates and triggers the process for that single state. That process should be generic (for ease of maintainance), so all state-specific information should be parametrizable.

There should be a database of whitelisted websites (per state), modifiable by the user. This can either be based on the endpoint, or on a separate access to the database. For simplicity, we'll take the first option. This database should contain all the state-specific parameters for regulation fetching, in such a way that a pipeline can be triggered by only specifying the state. This database helps satisfy FR-1.

The fetching process should save all websites visited in S3, and associate each document with its corresponding URL, as well as the time of retrieval. The easiest way of doing this is to keep a separate table where the S3 URI and all these metadata are stored, for later retrieval. That way, the URI works as an index for the metadata, satisfying FR-2, FR-5 and FR-10.

Considering the recoverability requirement, this requires that we design a graceful stop process, that either commits all information obtained through the process, up to the bundle, or avoids writing everything at all.

Also, Accuracy, Traceability, FR-5 and FR-10 require an external LLM logging system that guarantees total transparency about the workings of the fetching. This should be accomplished by a sidecar with a solution to that matter.